

Robert Weigel

professor

基本信息

姓名: Robert Weigel

所属院校: 埃朗根-纽伦堡大学

地区: Germany

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

通信工程

电子科学与技术

电磁场与无线技术

集成电路设计与集成系统

电子信息工程

电波传播与天线

研究领域

射频集成电路

微波与毫米波技术

无线通信与智能天线

模拟与射频IC设计

无线传感与射频技术

微波器件与电路

射频集成与系统设计

移动通信与5G/6G

微电子器件与制造

模拟与数模混合电路

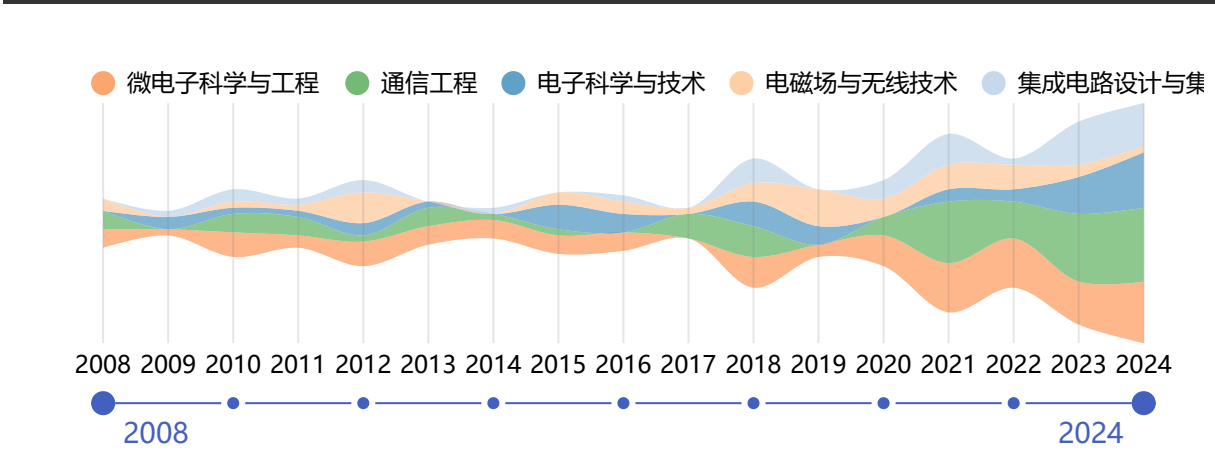
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个人简介

Robert Weigel was born in Germany in 1956. He holds Dr.-Ing. and Dr.-Ing.habil. degrees from the Munich University of Technology. He has had

various academic and industrial roles, including at Munich University of Technology, Siemens, and Tongji University. He co - founded DICE and LCM. Since 2002, he's been the Director of the Institute for Electronics Engineering at the University of Erlangen - Nuremberg. He has published over 550 papers, 2 books, etc., and supervised 75 PhD theses. He has won multiple awards, including the 2020 Microwave Career Award. He's an IEEE Fellow and a member of several professional associations. Contact: 49 - 9131 - 85 - 27200, Friedrich - Alexander Universitat, Erlangen - Nurnberg.

近年发表的论文



Lightweight and Person-Independent Radar-Based Hand Gesture Recognition for Classification and Regression of Continuous Gestures
IEEE INTERNET OF THINGS JOURNAL
Stadelmayer, Thomas; Hassab, Youcef; Servadei, Lorenzo; Santra, Avik; Weigel, Robert; Lurz, Fabian
被引用次数: 0 2024

Overview of Passive Radar and Its Receiver Architectures to Enhance Safety in Civil Aviation: A Comprehensive Analysis of History, Principles, and Performance Optimization Strategies
IEEE MICROWAVE MAGAZINE
Horlbeck, Marie; Scheiner, Benedict; Weigel, Robert; Lurz, Fabian
被引用次数: 1 2024

MTT-S Awards and the IMS2024 Awards Banquet
IEEE MICROWAVE MAGAZINE
Weigel, Robert
被引用次数: 0 2024

Hardware Design Challenges and Modulation Schemes in Joint Communication and Sensing: Analyzation and Comparison of the State-of-the-Art Co-Located Communication

and Sensing Systems

IEEE MICROWAVE MAGAZINE

Faghih-Naini, Samira; Sebastian, Caroline; Reissland, Torsten; Weigel, Robert; Scheiner, Benedict

被引用次数: 2

2024

Timed array architectures and integrated true-time delay elements for wideband millimeter-wave antenna arrays

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Koch, Manuel; Schoenhaerl, Stefan; Breun, Sascha; Fischer, Georg; Weigel, Robert

被引用次数: 0

2024

RT-SCNNs: real-time spiking convolutional neural networks for a novel hand gesture recognition using time-domain mm-wave radar data

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Shaaban, Ahmed; Strobel, Maximilian; Furtner, Wolfgang; Weigel, Robert; Lurz, Fabian

被引用次数: 1

2024

Volumetric vertical Hall element with integrated coil

SENSORS AND ACTUATORS A-PHYSICAL

Hohnloser, Daniel; Schmidt, Carsten; Spitzlsperger, Gerhard; Weigel, Robert

被引用次数: 0

2024

Resonate-and-Fire Spiking Neurons for Target Detection and Hand Gesture Recognition: A Hybrid Approach

2024 SMART SYSTEMS INTEGRATION CONFERENCE AND EXHIBITION, SSI 2024

Shaaban, Ahmed; Chaabouni, Zeineb; Strobel, Maximilian; Furtner, Wolfgang; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2024

High-Frequency Multiport Vector Network Analysis Methodology Using Port Extension Across Multiple VNAs

2024 54TH EUROPEAN MICROWAVE CONFERENCE, EUMC 2024

Loffler, Jens; Koch, Manuel; Breun, Sascha; Weigel, Robert

被引用次数: 0

2024

General Combiner Initialization Workflow for All Types of Active Load Modulation Power Amplifiers

PROCEEDINGS OF THE 2024 15TH GERMAN MICROWAVE CONFERENCE, GEMIC

Deublein, Alexander; Illmer, Andreas; Musolff, Christian; Weigel, Robert; Fischer, Georg

被引用次数: 0

2024

Evaluation of Control Link Impacts on the IRS Assisted Communication System with Hardware-in-the-Loop Simulation

PROCEEDINGS OF THE 2024 15TH GERMAN MICROWAVE CONFERENCE, GEMIC

Zeng, Xianshi; Buehlmeyer, Jonas; Weigel, Robert; Irnstorfer, Florian

被引用次数: 0

2024

Evaluating Synthetic Data Potential for 60 GHz FMCW Radar Simulations with Measurements

2024 INTERNATIONAL RADAR SYMPOSIUM, IRS 2024

Reitz, Philipp; Maiwald, Timo; Franchi, Norman; Weigel, Robert; Luebke, Maximilian

被引用次数: 1

2024

Performance Evaluation of a New Universal Time Domain Gating Algorithm

2024 IEEE INTERNATIONAL INSTRUMENTATION AND MEASUREMENT TECHNOLOGY CONFERENCE, I2MTC 2024

Zohner, Alina; Irnstorfer, Florian; Weigel, Robert; Pfannenmueller, Christof

被引用次数: 0

2024

A 290-359 GHz Push-Push Signal Source with 1.7 dBm P_{max} using Variable Inductance in SiGe BiCMOS Technology

2024 IEEE WIRELESS AND MICROWAVE TECHNOLOGY CONFERENCE, WAMICON

Breun, Sascha; Schrotz, Albert-Marcel; Koch, Manuel; Issakov, Vadim; Weigel, Robert

被引用次数: 0

2024

A 34 GHz CMOS VCO with Transformer Tail-Node Filter and TSPC Frequency Divider in 22 nm FDSOI

2024 IEEE 24TH TOPICAL MEETING ON SILICON MONOLITHIC INTEGRATED CIRCUITS IN RF SYSTEMS, SIRF

Engelmann, Andre; Probst, Florian; Hetterle, Philip; Weigel, Robert

被引用次数: 0

2024

A Broadband 22 nm FDSOI D-Band Power Amplifier with Dynamic Back Gate Bias Gain-Linearization Achieving 9.6% PAE at 8.7 dBm OP_{1dB} and 3.7% at 6 dB Back-off

2024 IEEE/MTT-S INTERNATIONAL MICROWAVE SYMPOSIUM, IMS 2024

Engelmann, Andre; Scheller, Kai; Probst, Florian; Koch, Manuel; Weigel, Robert; Fischer, Georg

被引用次数: 0

2024

Validation of a 4-Port and 3-Port Rat-Race Balun for the 60 GHz-Band

15TH IEEE LATIN AMERICAN SYMPOSIUM ON CIRCUITS AND SYSTEMS, LASCAS 2024

Faghih-Naini, Samira; Peters, Sebastian; Reissland, Torsten; Weigel, Robert

被引用次数: 0

2024

Deep Learning-based Person Detection on a Moving Robot

2024 IEEE TOPICAL CONFERENCE ON WIRELESS SENSORS AND SENSOR NETWORKS, WISNET

Gabsteiger, Jasmin; Maiwald, Timo; Kurin, Thomas; Dorn, Christian; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2024

Direct Sampling Receiver with an Adjustable Bandpass Filter for Use in Passive Radar with FM Radio

2024 IEEE RADIO AND WIRELESS SYMPOSIUM, RWS

Horlbeck, Marie; Fiedelak, Jonathan; Kurin, Thomas; Scheiner, Benedict; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2024

A Linearized Calibration Technique Using Modulated Signals for Wideband Dual-Input Doherty Characterization

2024 IEEE TOPICAL CONFERENCE ON RF/MICROWAVE POWER AMPLIFIERS FOR RADIO AND WIRELESS APPLICATIONS, PAWR

Illmer, Andreas; Deublein, Alexander; Weigel, Robert; Ackermann, Thomas

被引用次数: 0

2024

A 200-325 GHz Gain-Boosted J-Band Low-Noise Amplifier in a 130 nm SiGe BiCMOS Technology

2024 IEEE 24TH TOPICAL MEETING ON SILICON MONOLITHIC INTEGRATED CIRCUITS IN RF SYSTEMS, SIRF

Koch, Manuel; Breun, Sascha; Weigel, Robert

被引用次数: 0

2024

Gesture Recognition to Control a Moving Robot With FMCW Radar

2024 IEEE RADIO AND WIRELESS SYMPOSIUM, RWS

Maiwald, Timo; Gabsteiger, Jasmin; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2024

Improved Performance in PMCW Radar Systems Through Equalization Using Predistortion and Postprocessing

2024 IEEE/MTT-S INTERNATIONAL MICROWAVE SYMPOSIUM, IMS 2024

Peters, Sebastian; Faghih-Naini, Samira; Weigel, Robert; Reissland, Torsten

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2024

27 Gb/s PRBS Generator with In-Operation Programmable Taps for PMCW Radar

2024 IEEE 24TH TOPICAL MEETING ON SILICON MONOLITHIC INTEGRATED CIRCUITS IN RF SYSTEMS, SIRF

Probst, Florian; Engelmann, Andre; Weigel, Robert

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2024

A Digitally Controlled D-Band Phase Shifter for FMCW Radar and RadCom in SiGe BiCMOS

2024 IEEE WIRELESS AND MICROWAVE TECHNOLOGY CONFERENCE, WAMICON

Schrotz, Albert-Marcel; Breun, Sascha; Schoenhaerl, Stefan; Weigel, Robert

被引用次数: 0

2024

Resonate-and-Fire Spiking Neurons for Hand Gesture Label Refinement

2024 IEEE TOPICAL CONFERENCE ON WIRELESS SENSORS AND SENSOR NETWORKS, WISNET

Shaaban, Ahmed; Chaabouni, Zeineb; Strobel, Maximilian; Furtner, Wolfgang; Weigel, Robert; Lurz, Fabian

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2024

Analysis and compensation of frequency offset in co-site adjacent channel interference suppression

PROCEEDINGS OF THE IEEE

Maiwald, Tim; Li, Teng; Hotopan, George-Roberto; Kolb, Katharina; Disch, Karina; Potschka, Julian; Haag, Alexander; Dietz, Marco; Debaille, Bjorn; Zwick, Thomas; Aufinger, Klaus; Ferling, Dieter; Weigel, Robert; Visweswaran, Akshay

被引用次数: 21

2023

High-Accuracy True Speed-Over-Ground Measurement Approaches for Rail Vehicles: Comparison of Highly Accurate State of the Art True Speed-Over-Ground Measurement Approaches for Rail Vehicles

IEEE MICROWAVE MAGAZINE

Reissland, Torsten; Weigel, Robert; Koelpin, Alexander; Lurz, Fabian

被引用次数: 0

2023

A Microwave Rock Has Gone In Memoriam

IEEE MICROWAVE MAGAZINE

Boeck, Georg; Jacob, Arne; Vossiek, Martin; Weigel, Robert

被引用次数: 0

2023

MTT-S Awards and the IMS2023 Awards Banquet

IEEE MICROWAVE MAGAZINE

Weigel, Robert

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2023

A Direction-of-Arrival Estimation System for UAV-Assisted Search and Rescue: Locating Mobile Phones to Improve the Survival Chance of Disaster Victims

IEEE MICROWAVE MAGAZINE

Depold, Andreas; Dorn, Christian; Erhardt, Stefan; Weigel, Robert; Lurz, Fabian

被引用次数: 5

2023

Doherty's Dominance Continued: A Linear PA Using GaN-on-Si

IEEE MICROWAVE MAGAZINE

Deublein, Alexander; Loose, Michael; Illmer, Andreas; Musolff, Christian; Irnstorfer, Florian; Weigel, Robert; Ackermann, Thomas

被引用次数: 4

2023

Robust Permittivity Reconstruction With the Transmission and Reflection Method at mm-Wave Frequencies

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kornprobst, Simon; Engelsberger, Franz Xaver; Wojnowski, Maciej; Weigel, Robert

被引用次数: 1

2023

MOSFET-Based RF Switch With Active Biasing

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Solomko, Valentyn; Syroiezhin, Semen; Tayari, Danial; Essel, Jochen; Weigel, Robert

被引用次数: 0

2023

High-Voltage Analog Switch Inspired by Stacked MOSFET-Based RF Switches

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Solomko, Valentyn; Syroiezhin, Semen; Weigel, Robert

被引用次数: 1

2023

Mobile-BAT-A Novel Ultra-Low Power Wildlife Tracking System

SENSORS

Erhardt, Stefan; Koch, Martin; Kiefer, Andreas; Veith, Michael; Weigel, Robert; Koelpin, Alexander

被引用次数: 0

2023

A Wireless Joint Communication and Localization EMG-Sensing Concept for Movement Disorder Assessment

IEEE JOURNAL OF ELECTROMAGNETICS RF AND MICROWAVES IN MEDICINE AND BIOLOGY

Bruckner, Stefan; Kolpak, Jasmin; Michler, Fabian; Shanin, Nikita; Schober, Robert; Hagelauer, Amelie; Weigel, Robert; Gasner, Heiko; Winkler, Jurgen; Eskofier, Bjorn M.; Vossiek, Martin

被引用次数: 0

2023

A Synchronized 35 GHz Divide-by-5 TSPC Flip-Flop Clock Divider in 22 nm FDSOI

2023 ASIA-PACIFIC MICROWAVE CONFERENCE, APMC

Probst, Florian; Engelmann, Andre; Weigel, Robert

被引用次数: 2

2023

A Compact Energy-Saving D-Band Frequency Quadrupler with-1 dB Conversion Gain in 22 nm FDSOI

2023 ASIA-PACIFIC MICROWAVE CONFERENCE, APMC

Engelmann, Andre; Probst, Florian; Hetterle, Philip; Weigel, Robert

被引用次数: 0

2023

A Novel Noninvasive Micro-Doppler-Radar-Based Beehive Monitoring System

2023 ASIA-PACIFIC MICROWAVE CONFERENCE, APMC

Kurin, Thomas; Horlbeck, Marie; Dafner, Marion; Dorn, Christian; Reissland, Torsten; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2023

A 3-Stacked 10.2 dBm OP1dB D-Band Power Amplifier in 22 nm FDSOI for 6G Communication

2023 ASIA-PACIFIC MICROWAVE CONFERENCE, APMC

Huang, Tianzhu; Engelmann, Andre; Hetterle, Philip; Probst, Florian; Weigel, Robert

被引用次数: 1

2023

FPGA Based Evaluation Platform for High Performance Digital mmWave Radar Systems

2023 ASIA-PACIFIC MICROWAVE CONFERENCE, APMC

Peters, Sebastian; Faghih-Naini, Samira; Erhardt, Stefan; Weigel, Robert; Reissland, Torsten

被引用次数: 1

2023

Design of a Super Heterodyne Receiver with Double Conversion for Passive Radar with FM Radio

2023 ASIA-PACIFIC MICROWAVE CONFERENCE, APMC

Horlbeck, Marie; Kurin, Thomas; Scheiner, Benedict; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2023

A Low Voltage Folded Gilbert Cell Mixer for 6G Communication Systems

2023 ASIA-PACIFIC MICROWAVE CONFERENCE, APMC

Hetterle, Philip; Engelmann, Andre; Probst, Florian; Huang, Tianzhu; Weigel, Robert

被引用次数: 0

2023

Hybrid Sensing and Communications Receiver Front-End for Mobile Communications Applications

2023 IEEE MTT-S LATIN AMERICA MICROWAVE CONFERENCE, LAMC

Schuster, Armin; Reissland, Torsten; Weigel, Robert

被引用次数: 1

2023

A Continuously Adjustable True Time Delay for D-Band Timed Antenna Arrays

2023 18TH EUROPEAN MICROWAVE INTEGRATED CIRCUITS CONFERENCE, EUMIC

Koch, Manuel; Probst, Florian; Breun, Sascha; Weigel, Robert

被引用次数: 1

2023

Design of an E-TSPC Flip-Flop for a 43 Gb/s PRBS Generator in 22 nm FDSOI

2023 18TH EUROPEAN MICROWAVE INTEGRATED CIRCUITS CONFERENCE, EUMIC

Probst, Florian; Weninger, Jonas; Engelmann, Andre; Issakov, Vadim; Weigel, Robert

被引用次数: 1

2023

Sensor Resistance Based Sensitivity Temperature Drift Tracking of Integrated 3D Hall Sensors

2023 IEEE SENSORS

Krause, Dennis; Stahl-Offergeld, Markus; Sand, Markus; Kohlbrenner, Christian; Weigel, Robert

被引用次数: 1

2023

Dual-Mode Substrate Wave Cancellation in a 120 GHz 2 x 2 On-Chip Dipole Array

IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION

Lammert, Vincent; Leyrer, Michael; Hamouda, Mohamed; Weigel, Robert; Issakov, Vadim

被引用次数: 8

2022

The MTT-S Awards Committee-Evaluating the Excellence, Quality, and Service of Distinguished Microwave Engineers [AdCom Awards Committee]

IEEE MICROWAVE MAGAZINE

Weigel, Robert

被引用次数: 0

2022

Bringing Portability to a New Level

IEEE MICROWAVE MAGAZINE

Peters, Sebastian; Horlbeck, Marie; Probst, Florian; Weigel, Robert

被引用次数: 0

2022

MTT-S Awards and the IMS2022 Awards Banquet

IEEE MICROWAVE MAGAZINE

Weigel, Robert

被引用次数: 0

2022

Full Physical Layer Simulation Tool to Design Future 77 GHz JCRS-Applications

IEEE ACCESS

Luebke, Maximilian; Su, Yanpeng; Cherian, Athira J.; Fuchs, Jonas; Dubey, Anand; Weigel, Robert; Franchi, Norman

被引用次数: 7

2022

Radar-Based Gesture Recognition Using a Variational Autoencoder With Deep Statistical Metric Learning

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Stadelmayer, Thomas; Santra, Avik; Weigel, Robert; Lurz, Fabian

被引用次数: 6

2022

Investigation of Systematic Errors of an Indirect Resonant Planar Material Characterization Method

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Lau, Isabella; Weigel, Robert

被引用次数: 2

2022

Linearity Analysis of High-Voltage RF Switches for Antenna Tuning Applications

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Oezdamar, Oguzhan; Weigel, Robert; Hagelauer, Amelie; Solomko, Valentyn

被引用次数: 4

2022

Crosstalk in Elliptical Sensor Arrays for Current Measurement

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Zapf, Florian; Weiss, Roland; Weigel, Robert

被引用次数: 9

2022

Evaluation of Spiking Neural Networks for Time Domain-based Radar Hand Gesture Recognition

2022 23RD INTERNATIONAL RADAR SYMPOSIUM (IRS)

Shaaban, Ahmed; Furtner, Wolfgang; Weigel, Robert; Lurz, Fabian

被引用次数: 4

2022

Design of two Low DC-Power High-Efficiency D-Band Power Amplifiers in 22 nm FDSOI

2022 17TH EUROPEAN MICROWAVE INTEGRATED CIRCUITS CONFERENCE (EUMIC 2022)

Engelmann, Andre; Hetterle, Philip; Probst, Florian; Weigel, Robert; Dietz, Marco

被引用次数: 4

2022

Design of a Low Voltage D-band LNA in 22 nm FDSOI

2022 17TH EUROPEAN MICROWAVE INTEGRATED CIRCUITS CONFERENCE (EUMIC 2022)

Hetterle, Philip; Engelmann, Andre; Probst, Florian; Weigel, Robert; Dietz, Marco

被引用次数: 7

2022

Spiking Neural Networks for Gesture Recognition Using Time Domain Radar Data

2022 19TH EUROPEAN RADAR CONFERENCE (EURAD)

Shaaban, Ahmed; Furtner, Wolfgang; Weigel, Robert; Lurz, Fabian

被引用次数: 5

2022

Complex-Valued Neural Networks for Millimeter Wave FMCW-Radar Angle Estimations

2022 19TH EUROPEAN RADAR CONFERENCE (EURAD)

Kaiser, Kevin; Daugalas, Jonas; Lopez-Randulfe, Javier; Knoll, Alois; Weigel, Robert; Lurz, Fabian

被引用次数: 1

2022

Design of a High Linear, Frequency Selective VHF-Receiver with Low Phase Noise for a Passive Radar System

2022 19TH EUROPEAN RADAR CONFERENCE (EURAD)

Horlbeek, Marie; Erhardt, Stefan; Weigel, Robert; Lurz, Fabian

被引用次数: 2

2022

A 15-Gb/s PMCW Radar PRBS-Generator for MIMO and Joint Radar-Communication Systems

2022 ASIA-PACIFIC MICROWAVE CONFERENCE (APMC)

Probst, Florian; Engelmann, Andre; Hetterle, Philip; Issakov, Vadim; Weigel, Robert; Dietz, Marco

被引用次数: 8

2022

A Low-Voltage Broadband D-Band BPSK Modulator for a PMCW Radar Transmitter in 22 nm FDSOI

2022 ASIA-PACIFIC MICROWAVE CONFERENCE (APMC)

Engelmann, Andre; Probst, Florian; Hetterle, Philip; Weigel, Robert; Dietz, Marco

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2022

Cross-modal Learning of Graph Representations using Radar Point Cloud for Long-Range Gesture Recognition

2022 IEEE 12TH SENSOR ARRAY AND MULTICHANNEL SIGNAL PROCESSING WORKSHOP (SAM)

Hazra, Souvik; Feng, Hao; Kiprit, Gamze Naz; Stephan, Michael; Servadei, Lorenzo; Wille, Robert; Weigel, Robert; Santra, Avik

被引用次数: 4

2022

Circular Gysel Divider for the Frequency Range from 18GHz to 26GHz

2022 52ND EUROPEAN MICROWAVE CONFERENCE (EUMC)

Gabsteiger, Jasmin; Beck, Christopher; Dietz, Marco; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2022

Circular Gysel Divider for the Frequency Range from 18GHz to 26GHz

2022 52ND EUROPEAN MICROWAVE CONFERENCE (EUMC)

Gabsteiger, Jasmin; Beck, Christopher; Dietz, Marco; Weigel, Robert; Lurz, Fabian

被引用次数: 0

2022

A Bayesian Framework for Integrated Deep Metric Learning and Tracking of Vulnerable Road Users Using Automotive Radars

IEEE ACCESS

Dubey, Anand; Santra, Avik; Fuchs, Jonas; Luebke, Maximilian; Weigel, Robert; Lurz, Fabian

被引用次数: 18

2021

Millimeter-Wave and Terahertz Transceivers in SiGe BiCMOS Technologies

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Kissinger, Dietmar; Kahmen, Gerhard; Weigel, Robert

被引用次数: 47

2021

High-Resolution Millimeter-Wave Tomography System for Nondestructive Testing of Low-Permittivity Materials

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Och, Andreas; Hoelzl, Patrick A.; Schuster, Stefan; Scheiblhofer, Stefan; Zankl, Dominik; Pathuri-Bhuvana, Venkata; Weigel, Robert

被引用次数: 18

2021

Radar-based monitoring of glass fiber reinforced composites during fatigue testing

STRUCTURAL CONTROL & HEALTH MONITORING

Moll, Jochen; Maetz, Thomas; Maelzer, Moritz; Krozer, Viktor; Mischke, Kevin; Krause, Stefan; Bagemiel, Oliver; Nuber, Andreas; Kremling, Stefan; Kurin, Thomas; Lurz, Fabian; Weigel, Robert; Issakov, Vadim

被引用次数: 6

2021

A Broadband Zero-IF Down-Conversion Mixer in 130 nm SiGe BiCMOS for Beyond 5G Communication Systems in D-Band

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Maiwald, Tim; Potschka, Julian; Kolb, Katharina; Dietz, Marco; Hagelauer, Amelie; Visweswaran, Akshay; Weigel, Robert

被引用次数: 4

2021

Postprocessing and Evaluation for a Radar-Based True-Speed-Over-Ground Estimation System

IEEE MICROWAVE AND WIRELESS COMPONENTS LETTERS

Reissland, Torsten; Michler, Fabian; Scheiner, Benedict; Koelpin, Alexander; Weigel, Robert

被引用次数: 3

2021

Miniaturized Hybrid Frequency Reader for Contactless Measurement Scenarios Using Resonant Surface Acoustic Wave Sensors

SENSORS

Scheiner, Benedict; Probst, Florian; Michler, Fabian; Weigel, Robert; Koelpin, Alexander; Lurz, Fabian

被引用次数: 7

2021

A 10 kHz to 6 GHz Low-Cost Vector Network Analyzer

ADVANCES IN RADIO SCIENCE

Depold, Andreas; Erhardt, Stefan; Weigel, Robert; Lurz, Fabian

被引用次数: 1

2021

A Low-Power Squaring Circuit with Regulated Output and Improved Settling Time in 180 nm CMOS for 3-5 GHz IR-UWB Applications

ADVANCES IN RADIO SCIENCE

Schrufer, Daniel; Rober, Jurgen; Mai, Timo; Weigel, Robert

被引用次数: 0

2021

Resource Efficient Gesture Sensing Based on FMCW Radar using Spiking Neural Networks

2021 IEEE MTT-S INTERNATIONAL MICROWAVE SYMPOSIUM (IMS)

Arsalan, Muhammad; Chmurski, Mateusz; Santra, Avik; El-Masry, Moamen; Weigel, Robert; Issakov, Vadim

被引用次数: 8

2021

Design and measurements of a 28 GHz High-Linearity LNA in 45nm SOI-CMOS

2021 IEEE INTERNATIONAL CONFERENCE ON MICROWAVES, ANTENNAS, COMMUNICATIONS AND ELECTRONIC SYSTEMS (COMCAS)

Lammert, Vincent; Sakalas, Paulius; Werthof, Andreas; Weigel, Robert; Issakov, Vadim

被引用次数: 3

2021

Hybrid C-Tuner IC for 40V/80V Antenna Aperture Tuning Applications

2021 IEEE INTERNATIONAL CONFERENCE ON MICROWAVES, ANTENNAS, COMMUNICATIONS AND ELECTRONIC SYSTEMS (COMCAS)

Oezdamar, Oguzhan; Syroiezhin, Semen; Weigel, Robert; Hagelauer, Amelie; Solomko, Valentyn

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2021

Considerations for Capacitance Measurements of Antenna Tuning RF Switches on Board

2021 IEEE INTERNATIONAL CONFERENCE ON MICROWAVES, ANTENNAS, COMMUNICATIONS AND ELECTRONIC SYSTEMS (COMCAS)

Solomko, Valentyn; Oezdamar, Oguzhan; Tayari, Danial; Voelkel, Matthias; Weiss, Simon; Essel, Jochen; Weigel, Robert; Hagelauer, Amelie

被引用次数: 0

2021

Towards Broadband D-Band Wireless Communication Systems using Beam Steering in SiGe BiCMOS Technology

2021 13TH GLOBAL SYMPOSIUM ON MILLIMETER-WAVES & TERAHERTZ (GSMM)

Maiwald, Tim; Kolb, Katharina; Potschka, Julian; Dietz, Marco; Weigel, Robert

被引用次数: 3

2021

A Bidirectional 28 GHz RF Transceiver Front-End with Test and Calibration Interface for 5G Phased Arrays

2021 16TH EUROPEAN MICROWAVE INTEGRATED CIRCUITS CONFERENCE (EUMIC 2021)

Kob, Katharina; Potschka, Julian; Maiwald, Tim; Aufinger, Klaus; Hagelauer, Amelie; Dietz, Marco; Weigel, Robert

被引用次数: 1

2021

Implementation and Assessment of a Radar Based True-Speed-Over-Ground Estimation Approach Utilizing Complex-Valued Correlation

2021 18TH EUROPEAN RADAR CONFERENCE (EURAD)

Reissland, Torsten; Michler, Fabian; Weigel, Robert; Koelpin, Alexander; Lurz, Fabian

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Analysis of a Physically-Embedded Radar Sensor System

2021 18TH EUROPEAN RADAR CONFERENCE (EURAD)

Kurin, Thomas; Issakov, Vadim; Erhardt, Stefan; Weigel, Robert; Lurz, Fabian

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A 9-Channel Phase Coherent Receive System for Direction of Arrival Estimation

2021 51ST EUROPEAN MICROWAVE CONFERENCE (EUMC)

Depold, Andreas; Dorn, Christian; Erhardt, Stefan; Weigel, Robert; Lurz, Fabian

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Accurate Power Loss Measurements of Aperture Tuning RF Switches on Board

2021 IEEE ASIA-PACIFIC MICROWAVE CONFERENCE (APMC)

Solomko, Valentyn; Oezdamar, Oguzhan; Rimmelspacher, Johannes; Nascimento, Pablo; Muerke, Michael; Pyo, Chul Seung; Weigel, Robert; Hagelauer, Amelie

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Quad-Band Inverted-F Antenna System Tuned by Hybrid C-Tuner

2021 IEEE ASIA-PACIFIC MICROWAVE CONFERENCE (APMC)

Oezdamar, Oguzhan; Solomko, Valentyn; Weigel, Robert; Hagelauer, Amelie

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Model of Substrate Capacitance of MOSFET RF Switch Inspired by Inverted Microstrip Line

IEEE 51ST EUROPEAN SOLID-STATE DEVICE RESEARCH CONFERENCE (ESSDERC 2021)

Solomko, Valentyn; Oezdamar, Oguzhan; Weigel, Robert; Hagelauer, Amelie

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Validation and Analysis of the Propagation Channel at 60 GHz for Vehicular Communication

2021 IEEE 94TH VEHICULAR TECHNOLOGY CONFERENCE (VTC2021-FALL)

Luebke, Maximilian; Fuchs, Jonas; Dubey, Anand; Hamoud, Hussein; Dressler, Falko; Weigel, Robert; Lurz, Fabian

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Improved Target Detection and Feature Extraction using a Complex-Valued Adaptive Sinc Filter on Radar Time Domain Data

29TH EUROPEAN SIGNAL PROCESSING CONFERENCE (EUSIPCO 2021)

Stadelmayer, Thomas; Santra, Avik; Stadelmayer, Markus; Weigel, Robert; Lurz, Fabian

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People Counting Solution Using an FMCW Radar with Knowledge Distillation From Camera Data

2021 IEEE SENSORS

Stephan, Michael; Hazra, Souvik; Santra, Avik; Weigel, Robert; Fischer, Georg

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Reconfigurable Inverted-F Antenna for MIMO Cellular User Equipment

2021 IEEE-APS TOPICAL CONFERENCE ON ANTENNAS AND PROPAGATION IN WIRELESS COMMUNICATIONS (APWC)

Oezdamar, Oguzhan; Weigel, Robert; Hagelauer, Amelie; Wang, Kun; Solomko, Valentyn

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Large-Signal Performance Optimization of Inverted-F Antenna Tuned by Switchable RF Capacitor

2021 IEEE-APS TOPICAL CONFERENCE ON ANTENNAS AND PROPAGATION IN WIRELESS COMMUNICATIONS (APWC)

Solomko, Valentyn; Oezdamar, Oguzhan; Weigel, Robert; Hagelauer, Amelie

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A Robust Digital Predistortion Algorithm for 5G MIMO: Modeling a MIMO Scenario With Two Nonlinear MIMO Transmitters Including a Cross-Coupling Effect

IEEE MICROWAVE MAGAZINE

Ackermann, Thomas; Potschka, Julian; Maiwald, Tim; Hagelauer, Amelie; Fischer, Georg; Weigel, Robert

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2020

Nothing Beats SNR: Single-Digit Micrometer Ranging Using a Low-Power CW Radar Featuring a Low-Weight 3D-Printed Horn Antenna

IEEE MICROWAVE MAGAZINE

Scheiner, Benedict; Michler, Fabian; Lurz, Fabian; Weigel, Robert; Koelpin, Alexander

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2020

Considerations for Harmonics Distribution in Aperture-Tuned Inverted-F Antenna for Cellular Handheld Devices

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Oezdamar, Oguzhan; Weigel, Robert; Hagelauer, Amelie; Solomko, Valentyn

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2020

Low-Weight Wireless Sensor Node With Sensor-Data-Enhanced Dual-Frequency RSSI-Based Distance Estimation

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Duda, Niklas; Weigel, Robert; Koelpin, Alexander

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2020

Multilayer Topology Optimization of Wideband SIW-to-Waveguide Transitions

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Hassan, Emadeldeen; Scheiner, Benedict; Michler, Fabian; Berggren, Martin; Wadbro, Eddie; Roehrl, Franz; Zorn, Stefan; Weigel, Robert; Lurz, Fabian

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Design of a 60 GHz 32% PAE Class-AB PA with 2nd Harmonic Control in 45-nm PD-SOI CMOS

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ciocoveanu, Radu; Weigel, Robert; Hagelauer, Amelie; Issakov, Vadim

被引用次数: 6

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A 130 GHz Fully-Integrated Fundamental-Frequency D-Band Transmitter Module With > 4 dBm Single-Ended Output Power

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Aguilar, Erick; Issakov, Vadim; Weigel, Robert

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2020

Mechanically Flexible Sensor Array for Current Measurement

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Zapf, Florian; Weiss, Roland; Itzke, Alexander; Gordon, Renee; Weigel, Robert

被引用次数: 15

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Electrothermal Modeling of Surface Acoustic Wave Resonators and Filters

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Akstaller, Wolfgang; Weigel, Robert; Hagelauer, Amelie

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Delay analysis of high-electron mobility transistors under high drain bias

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Reiser, Korbinian; Twynam, John; Brech, Helmut; Weigel, Robert

被引用次数: 0

2020

A dataset of clinically recorded radar vital signs with synchronised reference sensor signals

SCIENTIFIC DATA

Schellenberger, Sven; Shi, Kilin; Steigleder, Tobias; Malessa, Anke; Michler, Fabian; Hameyer, Laura; Neumann, Nina; Lurz, Fabian; Weigel, Robert; Ostgathe, Christoph; Koelpin, Alexander

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A dataset of radar-recorded heart sounds and vital signs including synchronised reference sensor signals

SCIENTIFIC DATA

Shi, Kilin; Schellenberger, Sven; Will, Christoph; Steigleder, Tobias; Michler, Fabian; Fuchs, Jonas; Weigel, Robert; Ostgathe, Christoph; Koelpin, Alexander

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Continuous In-Bed Monitoring of Vital Signs Using a Multi Radar Setup for Freely Moving Patients

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被引用次数: 15

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Electronically Reconfigurable 6 x 6 Element Transmitarray at K-Band Based on Unit Cells With Continuous Phase Range

IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS

Frank, Martin; Lurz, Fabian; Weigel, Robert; Koelpin, Alexander

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2019

Enhancing RF Bulk Acoustic Wave Devices

IEEE MICROWAVE MAGAZINE

Chauhan, Vikrant; Huck, Christian; Frank, Astrid; Akstaller, Wolfgang; Weigel, Robert; Hagelauer, Amelie

被引用次数: 14

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(Micro)Metering with Microwaves

IEEE MICROWAVE MAGAZINE

Michler, Fabian; Scheiner, Benedict; Lurz, Fabian; Weigel, Robert; Koelpin, Alexander

被引用次数: 8

2019

Evaluation of a robust correlation-based true-speed-over-ground measurement system employing a FMCW radar

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Reissland, Torsten; Lenhart, Bjoern; Lichtblau, Johann; Sporer, Michael; Weigel, Robert; Koelpin, Alexander

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Bare die connections via aerosol jet technology for millimeter wave applications

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Roehrl, Franz Xaver; Jakob, Johannes; Bogner, Werner; Weigel, Robert; Zorn, Stefan

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Compact low-cost substrate integrated waveguide fed antenna for 122 GHz radar applications

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

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Compact polarimetry for automotive applications

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

Erhart, Christian; Lutz, Steffen; Mutschlerl, Marc A.; Scharf, Philipp A.; Walter, Thomas; Mantz, Hubert; Weigel, Robert

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Squirrel-Based Calibration Algorithm for Six-Port Radar

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Linz, Sarah; Lurz, Fabian; Weigel, Robert; Koelpin, Alexander

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Influence of the PCB Manufacturing Process on the Measurement Error of Planar Relative Permittivity Sensors Up To 100 GHz

IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES

Lau, Isabella; Hajian, Ali; Michler, Fabian; Gold, Gerald; Lurz, Fabian; Schmid, Ulrich; Helmreich, Klaus; Weigel, Robert; Koelpin, Alexander

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Simultaneous Inductive Power and Data Transmission System for Smart Applications

IEEE CIRCUITS AND SYSTEMS MAGAZINE

Trautmann, Martin; Sanftl, Benedikt; Weigel, Robert; Koelpin, Alexander

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Modified Gilbert-Cell Mixer With an LO Waveform Shaper and Switched Gate-Biasing for 1/f Noise Reduction in 28-nm CMOS

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Low-Cost Transmitarray Antenna Designs in V-Band based on Unit-Cells with 1 Bit Phase Resolution

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Accuracy Bounds and Measurements of a Contactless Permittivity Sensor for Gases Using Synchronized Low-Cost mm-Wave Frequency Modulated Continuous Wave Radar Transceivers

SENSORS

Och, Andreas; Schrattenecker, Jochen O.; Schuster, Stefan; Hoelzl, Patrick A.; Freidl, Philipp F.; Scheiblhofer, Stefan; Zankl, Dominik; Weigel, Robert

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A Clinically Evaluated Interferometric Continuous-Wave Radar System for the Contactless Measurement of Human Vital Parameters

SENSORS

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Fusion of Nonintrusive Environmental Sensors for Occupancy Detection in Smart Homes

IEEE INTERNET OF THINGS JOURNAL

Zimmermann, Lars; Weigel, Robert; Fischer, Georg

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Microw(h)att?! Ultralow-Power Six-Port Radar

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A flexible mixed-signal image processing pipeline using 3D chip stacks

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A 52-to-67 GHz dual-core push-push VCO in 40-nm CMOS

INTERNATIONAL JOURNAL OF MICROWAVE AND WIRELESS TECHNOLOGIES

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A 1 to 32 GHz broadband multi-octave receiver for monolithic integrated vector network analyzers in SiGe technology

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Dietz, Marco; Bauch, Andreas; Aufinger, Klaus; Weigel, Robert; Hagelauer, Amelie

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Antenna and Package Design for 61-and 122-GHz Radar Sensors in Embedded Wafer-Level Ball Grid Array Technology

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A Highly Reconfigurable RF-DPLL Phase Modulator for Polar Transmitters in Cellular RFICs

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Buckel, Tobias; Tertinek, Stefan; Mayer, Thomas; Bauernfeind, Thomas; Wicpalek, Christian; Springer, Andreas; Weigel, Robert; Ussmueller, Thomas

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High-Precision Interferometric Radar for Sheet Thickness Monitoring

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A Novel Digital-Intensive Hybrid Polar-I/Q RF Transmitter Architecture

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A Mixed-Signal Technique for TX-Induced Modulated Spur Cancellation in LTE-CA Receivers

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Elmaghraby, Ahmed; Kanumalli, Ram Sunil; Schelmbauer, Werner; Mayer, Andreas; Herzinger, Stefan; Schwartz, Daniel; Huemer, Mario; Weigel, Robert

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Voltage and Current Selector-Based Biasing Topology for Multiple Supply Voltage Circuits

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Easwaran, Navaneeth; Chellamuthu, Shanmuganand; Venugopal, Sunil Kashyap; Weigel, Robert

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Investigation on Third-Order Intermodulation Distortions Due to Material Nonlinearities in TC-SAW Devices

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BATS: Adaptive Ultra Low Power Sensor Network for Animal Tracking

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Reader Architectures for Wireless Surface Acoustic Wave Sensors

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Sporer, Michael; Weigel, Robert; Koelpin, Alexander

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X-Parameter Characterization of TC SAW Filters With Enhanced Dynamic Range

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A System Concept for Online Calibration of Massive MIMO Transceiver Arrays for Communication and Localization

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Highly Integrated 4-32-GHz Two-Port Vector Network Analyzers for Instrumentation and Biomedical Applications

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Nehring, Johannes; Schuetz, Martin; Dietz, Marco; Nasr, Ismail; Aufinger, Klaus; Weigel, Robert; Kissinger, Dietmar

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A method for the determination of the complex permittivity by detuned ring resonators for bulk materials up to 110 GHz

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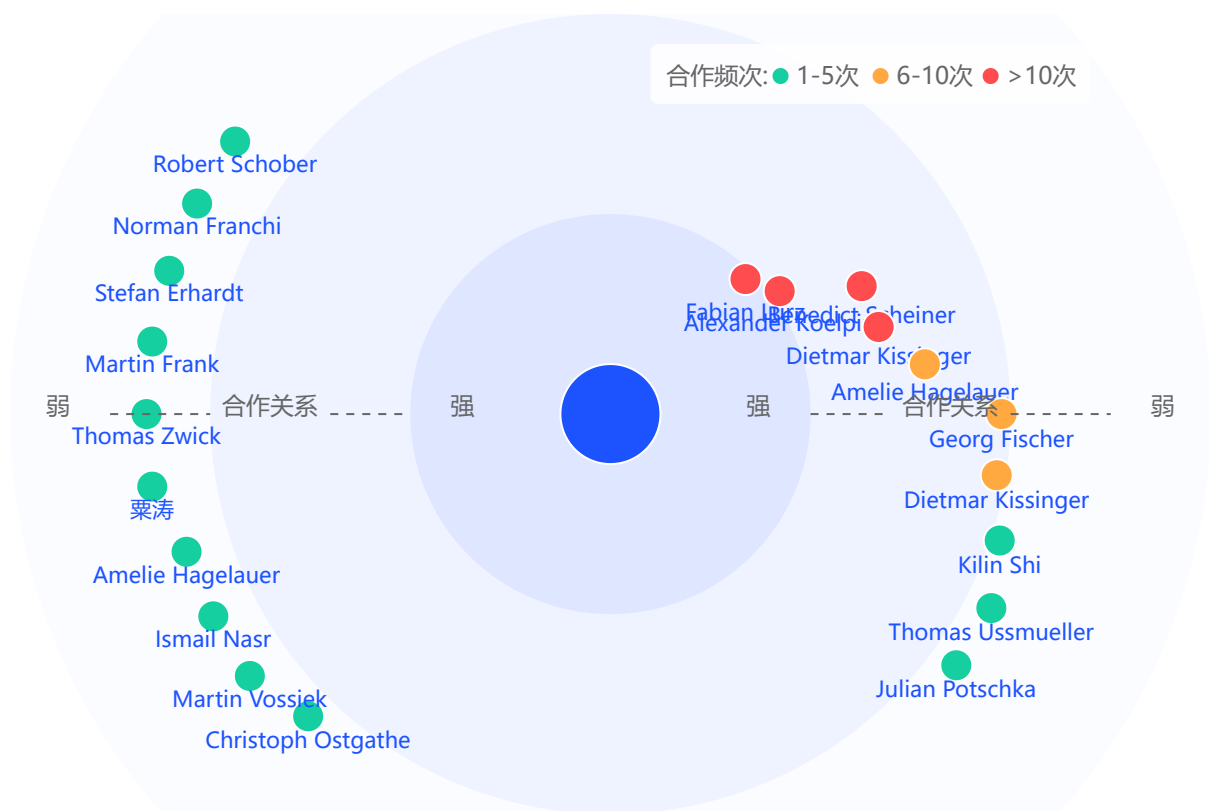
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